Guiv Farmanfarmaian

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Citizenship: US, Swiss, Canadian
Languages: English, French, Persian, Italian (C2), German (C2), Russian (A2), Spanish (A2)

PROFILE

Machine Learning researcher at ETH Zürich focused on stable **RL post-training for LLM reasoning**, with hands-on experience in RLVR (RL with Verifiable Rewards), test-time compute, and distributed training. Master's thesis develops new methods for preventing mode collapse and recovering learning signal on hard mathematical reasoning problems, with selected results under confidentiality pending publication. Strong mathematical foundation (**BSc Mathematics, ETH Zürich**), with a research agenda centered on **RL for reasoning.**

EDUCATION

ETH Zürich — MSc Computer Science (Machine Intelligence) *Zürich, Switzerland*
Sep 2023 – Feb 2026 • Major: Machine Intelligence • Minor: Theoretical Computer Science • GPA: 5.7/6.0 (top 10%

- **Master's Thesis:** *Improving LLM Reasoning on Challenging Problems* — supervised by Dr. Amir Joudaki and Prof. Dr. Thomas Hofmann; **manuscript in preparation.**
- **Semester Thesis:** *Improving Investment Strategies with GNNs* — supervised by Florian Grötschla, Joel Mathys, and Prof. Dr. Roger Wattenhofer (in collaboration with a leading investment firm; under confidentiality).
- **Selected coursework:** Deep Learning, Large Language Models, Machine Perception, Probabilistic Artificial Intelligence, Advanced Algorithms, Convex Optimization, Information Theory.

ETH Zürich — BSc Mathematics *Zürich, Switzerland*
Sep 2020 – Sep 2023 • Focus: Number Theory, Theoretical Computer Science

- **Selected coursework:** Probability Theory, Number Theory I & II, Algorithms / Probability / Computing (APC).

RESEARCH EXPERIENCE

Master's Thesis — RL Post-training & Test-Time Compute for Math Reasoning *Zürich, Switzerland*
ETH Zürich — Data Analytics Lab (Hofmann Group) • June 2025 – Feb 2026

- **Stable RLVR via teacher-hint conditioning.** Designed a GRPO variant that prevents mode collapse during RL post-training: **+54% policy entropy** and **~9× lower KL divergence** vs. the GRPO baseline, with **+3 pp on unhinted MATH-500 pass@8*. Confidential; manuscript in preparation***
- **Recovering RL signal on hard problems.** Developed a minimal-prefix solution-conditioning method that restores learning signal on problems with zero pass@16, yielding **+5 pp on the unhinted target benchmark** , addressing the well-known cold-start problem in RLVR.
- **Test-time compute methods.** Implemented and evaluated recursive reasoning, self-refinement, and best-of-N inference strategies on hard math problems; analyzed where test-time gains plateau and how they could be distilled into base weights.
- **Engineering.** Built a distributed RL training pipeline in PyTorch with vLLM-based rollouts, profiled and optimized GPU utilization on multi-node clusters.

Semester Thesis — Graph Learning for Financial Forecasting *Zürich, Switzerland*
ETH Zürich — Wattenhofer Group, in collaboration with a leading investment firm • Mar 2024 – Sep 2024

- Built and evaluated GNN and Graph Transformer architectures on a large directed supply-chain graph for predicting portfolio-relevant financial metrics.
- Ran controlled ablations comparing graph-based models to non-relational baselines, isolating the contribution of supply-chain topology to predictive accuracy and downstream portfolio performance.

RESEARCH INTERESTS

- **Stable RL for reasoning.** Why does RLVR collapse, and what's the right inductive bias to keep policies exploratory while still optimizing reward? Particularly interested in entropy regularization, off-policy correction, and reference-policy conditioning.
- **Sample-efficient reasoning.** Reaching strong reasoning performance with substantially less data. Fewer environment interactions also means greater control over what skills the model acquires and from where, making training more steerable and auditable.
- **Distilling test-time compute into weights.** Test-time recursion and best-of-N improve performance dramatically, but inference cost grows. How can these gains be amortized into the base policy through self-distillation or expert iteration?
- **Tiny Recursive Models for symbolic reasoning.** Extending TRM-style architectures beyond ARC toward algebraic and symbolic reasoning where verifiable rewards are tractable.
- **Continual learning.** Architectures and algorithms that mitigate catastrophic forgetting and enable models to keep improving from new data without re-training from scratch.

PROFESSIONAL EXPERIENCE

Teaching Assistant — Computational Intelligence Lab (CIL) *Zürich, Switzerland*
ETH Zürich • Feb 2025 – Jun 2025

- Designed and graded coursework, ran exercise sessions, and held office hours covering matrix factorization, deep generative models, and modern representation learning.

Funding Circle Ltd. — Data Scientist *London, United Kingdom*
Nov 2019 – Aug 2020

- Built regression and clustering models to analyze loan-portfolio risk; deployed Random Forest classifiers for early default detection during company hackathons.
- Automated reporting pipelines in Python and R; built data ingestion and processing on AWS.

Reply Sytel — IT Consulting Intern *London, United Kingdom*
Apr 2017 – May 2017

- Wrote full-stack software prototypes and authored white papers on next-generation networking and Proof-of-Stake consensus.

TECHNICAL SKILLS

ML Research: PyTorch, vLLM, Hugging Face Transformers, TRL, DeepSpeed, distributed training (FSDP / DDP), W&B, Slurm, GraphGPS, PyG.

Programming: Python, C++, TypeScript, R, Java.

Mathematics: Number theory, probability theory, convex optimization, theoretical CS.